QP CODE: 25901689

Answer Key with Mark Distribution

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCEEXAMINATION

APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCGEO100 - EXOGENOUSEARTH PROCESSES

(2024 ADMISSION ONWARDS)

MG2DSCGEO100 – Exogenous Earth Processes

Maximum Marks: 50

Part A – Multiple Choice Questions

(1 mark each)

| 1 | b) Condensation | Remember (K) | CO1 | 1 |

| 2 | c) In the lower part of the glacier between the interface with the bedrock and the overlying zone of rigidity | Understand (U) | CO3 | 1 |

| 3 | c) Waves carry sediments to different locations along the coast | Understand (U) | CO3 | 1 |

| 4 | a) Chemical weathering | Understand (U) | CO3 | 1 |

| 5 | b) Karst | Understand (U) | CO3 | 1 |

Total = 5 marks

Part B – Short Answer

(Answer any 7 questions – 3 marks each)

6. Explain the main types of erosion and how do they contribute to shaping the Earth's surface?

Understand (U) **Course Outcome:** CO3

Mark Distribution

- Definition of erosion – 1 mark
- Geological agents (water, wind, glacier, wave erosion etc.) Attrition, Abrasion, Rock plucking, Deflation – 1.5 mark
- Role in shaping landforms with examples – 0.5 mark

Key Points

- Erosion is the removal and transportation of weathered materials.
- Running water forms valleys and canyons.
- Wind creates dunes and mushroom rocks.
- Glaciers carve U-shaped valleys.
- Waves form cliffs and beaches.

7. Describe gully erosion.

Remember (K) **Course Outcome:** CO3

Mark Distribution

- Definition – 1 mark
- Process/mechanism – 1 mark
- Effects/examples/control measures – 1 mark

Key Points

- Gully erosion is removal of soil and weathered rock by concentrated runoff forming deep channels. Arid -semi arid climate and heavy rainfall are the triggering agents.
- Common in deforested or overgrazed areas.
- Leads to land degradation and loss of fertile soil, Bad land topography

8. Explain why wind erosion is more effective in arid regions than in humid regions.

Understand (U) **Course Outcome:** CO3

Mark Distribution

- Mention arid climatic conditions – 1 mark
- Lack of vegetation and dry loose sediments – 1 mark
- Strong wind action and limited moisture binding – 1 mark

Key Points

- Sparse vegetation exposes soil.
- Dry particles are easily transported.
- High wind velocity and lack of moisture enhance erosion.

9. Differentiate between abrasion and attrition.

Understand (U) **Course Outcome:** CO3

Mark Distribution

- Definition of abrasion – 1 mark
- Definition of attrition – 1 mark
- Comparative distinction/example – 1 mark

Key Points

Abrasion	**Attrition**
Wearing away of surface by friction	Collision among transported particles
Acts on bedrock	Acts on sediments/rock fragments
Produces polished surfaces	Produces smaller rounded particles

10. Describe the role played by abrasion in forming features like cliffs and caves.

Understand (U) **Course Outcome:** CO3

Mark Distribution

- Definition of abrasion – 1 mark
- Explanation of wave action/mechanical wearing – 1 mark
- Formation of cliffs/caves – 1 mark

Key Points

- Waves carrying sediments erode coastal rocks.
- Continuous abrasion undercuts cliffs.
- Weak zones enlarge into sea caves.

11. Describe the different types of coastal processes that shape a coastline.

Understand (U) **Course Outcome:** CO3

Mark Distribution

- Mention erosional processes – 1 mark
- Mention transportation processes – 1 mark
- Mention depositional processes with examples – 1 mark

Key Points

- Erosion: hydraulic action, abrasion.
- Transportation: longshore drift.
- Deposition forms beaches, spits, bars.

12. Describe the role of sea water as a geomorphic agent.

Understand (U) **Course Outcome:** CO3

Mark Distribution

- Sea water as an erosional agent – 1 mark
- Transportation role – 1 mark
- Depositional role/landforms – 1 mark

Key Points

- Waves erode coasts.
- Currents transport sediments.
- Deposition forms beaches and coastal plains.

13. Describe the importance of the atmosphere in the Earth system.

Understand (U) **Course Outcome:** CO1

Mark Distribution

- Composition/functions of atmosphere – 1 mark
- Regulation of climate and weather – 1 mark
- Support for life/protection role – 1 mark

Key Points

- Provides gases essential for life.
- Controls temperature and hydrologic cycle.
- Protects Earth from harmful radiation.

14. Explain how wind abrasion affects rock surfaces in desert environments.

Understand (U) **Course Outcome:** CO3

Mark Distribution

- Definition of wind abrasion – 1 mark
- Process of sand blasting – 1 mark
- Resulting landforms/features – 1 mark

Key Points

- Wind-driven sand particles erode rock surfaces.

- Produces grooves, polish, ventifacts and mushroom rocks.

Part C – Short Notes

(Answer any 3 questions – 8 marks each)

15. Analyze the role of human activities in accelerating or hindering weathering.

Understand (U) **Course Outcome:** CO2

Mark Distribution

- Introduction to weathering – 1 mark
- Human activities accelerating weathering – 3 marks
- Human activities reducing weathering – 2 marks
- Examples/conclusion – 2 marks

Key Points

- Mining, quarrying, pollution and deforestation accelerate weathering.
- Acid rain enhances chemical weathering.
- Afforestation and soil conservation reduce weathering.
- Urbanization exposes rocks to physical and chemical changes.

16. Glacier is a powerful geomorphic agent to form various landforms through erosion. Do you agree? Substantiate your views.

Understand (U) **Course Outcome:** CO3

Mark Distribution

- Introduction to glaciers – 1 mark
- Mechanisms of glacial erosion (plucking, abrasion) – 2 marks
- Erosional landforms – 3 marks
- Conclusion with evaluation – 2 marks

Key Points

- Glaciers erode through plucking and abrasion.
- Form cirques, arêtes, horns, U-shaped valleys and fjords.
- Glaciers transport large debris and reshape landscapes extensively.

17. Discuss how meandering streams differ from braided streams. Add a note on formation of an oxbow lake.

Understand (U) **Course Outcome:** CO3

Mark Distribution

- Characteristics of meandering streams – Stage of fluvial evolution-2 marks
- Characteristics of braided streams –Sediment load and transportation failure- 2 marks
- Differences between them – 2 marks
- Formation of oxbow lake – 2 marks

Key Points

Meandering Streams	**Braided Streams**
Single sinuous channel	Multiple interwoven channels
Gentle slope	Steeper slope
Fine sediments	Coarse sediments
Stable discharge	Variable discharge

Oxbow Lake Formation

- Meander loops enlarge due to lateral erosion.
- Neck becomes narrow.
- River cuts through during flood.
- Abandoned meander becomes oxbow lake.

18. Discuss how waves and currents contribute to the deposition of sediments along coastlines.

Understand (U) **Course Outcome:** CO3

Mark Distribution

- Role of waves – 2 marks
- Role of currents/longshore drift – 2 marks
- Depositional landforms – 2 marks
- Examples/conclusion – 2 marks

Key Points

- Constructive waves deposit sediments.
- Longshore currents transport and redistribute materials.
- Formation of beaches, spits, bars and tombolos.

19. Discuss in detail the various types of weathering products and their significance in soil formation.

Understand (U) **Course Outcome:** CO2

Mark Distribution

- Introduction to weathering products – 1 mark
- Physical weathering products – 2 marks
- Chemical weathering products – 3 marks
- Role in soil formation/significance – 2 marks

Key Points

- Physical weathering produces rock fragments and regolith.
- Chemical weathering forms clay minerals, oxides and dissolved ions.
- Organic matter and degradation
- Weathering products contribute to soil texture, fertility and horizon development.

****